

Ex.No: 1	Install VM Workstation in Ubuntu and set up windows and kali.
Date:	

AIM:

To install VMware Workstation and create virtual machines for Ubuntu and Kali Linux, and configure both operating systems inside VMware Workstation.

Procedure:

Step 1: Download VMware Workstation

1. Open a web browser on the host computer.
2. Visit the official VMware/Broadcom download page.
3. Download VMware Workstation Pro according to the host operating system. Official VMware Workstation download:

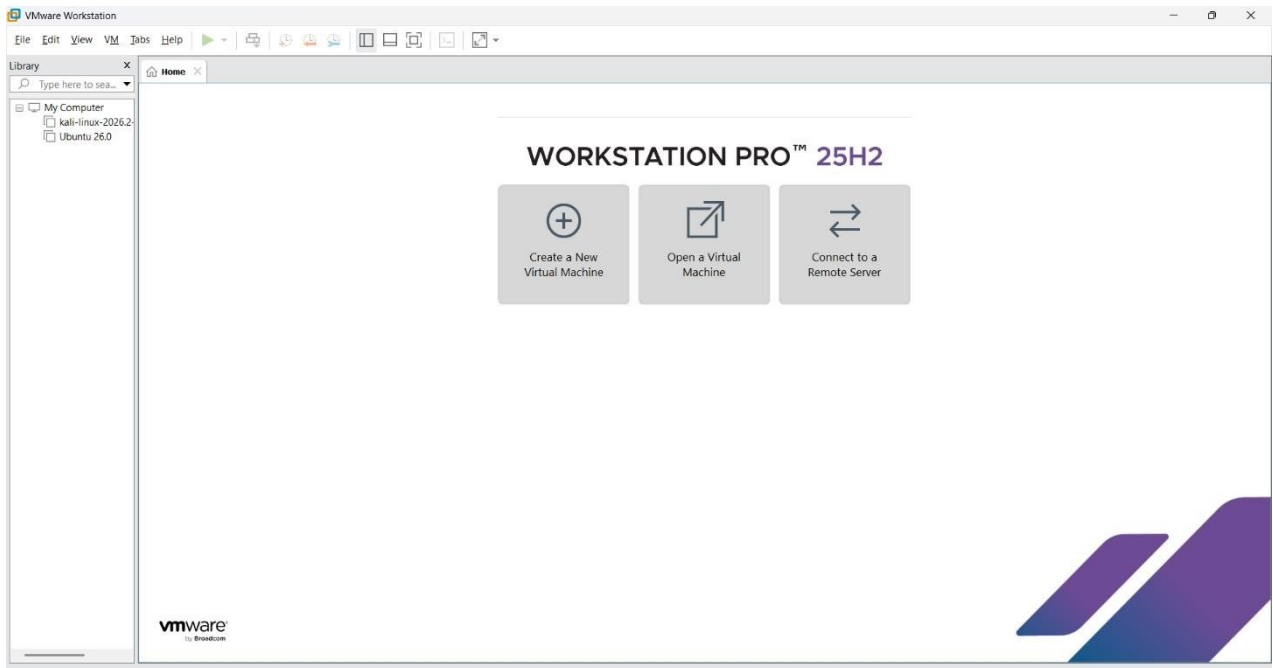
[VMware Workstation Pro – Broadcom Official Download](#)

4. Complete the download and keep the installer in an easily accessible folder.

Step 2: Install VMware Workstation

1. Locate the downloaded VMware Workstation installer.
2. Run the installer as administrator.
3. Follow the installation wizard.
4. Accept the license agreement.
5. Select the required installation options.
6. Complete the installation.
7. Restart the host computer if requested.

8. Open VMware Workstation after installation.



Step 3: Prepare the Ubuntu ISO

1. Download the Ubuntu ISO image from the official Ubuntu website.
2. Select the appropriate 64-bit Desktop ISO.
3. Save the ISO file on the host computer.
4. The ISO file will be used as the installation media for the Ubuntu virtual machine. Official Ubuntu download:

[Ubuntu Desktop Download](#)

Step 4: Create the Ubuntu Virtual Machine

1. Open VMware Workstation.
2. Select **Create a New Virtual Machine**.
3. Select **Typical (Recommended)** configuration.

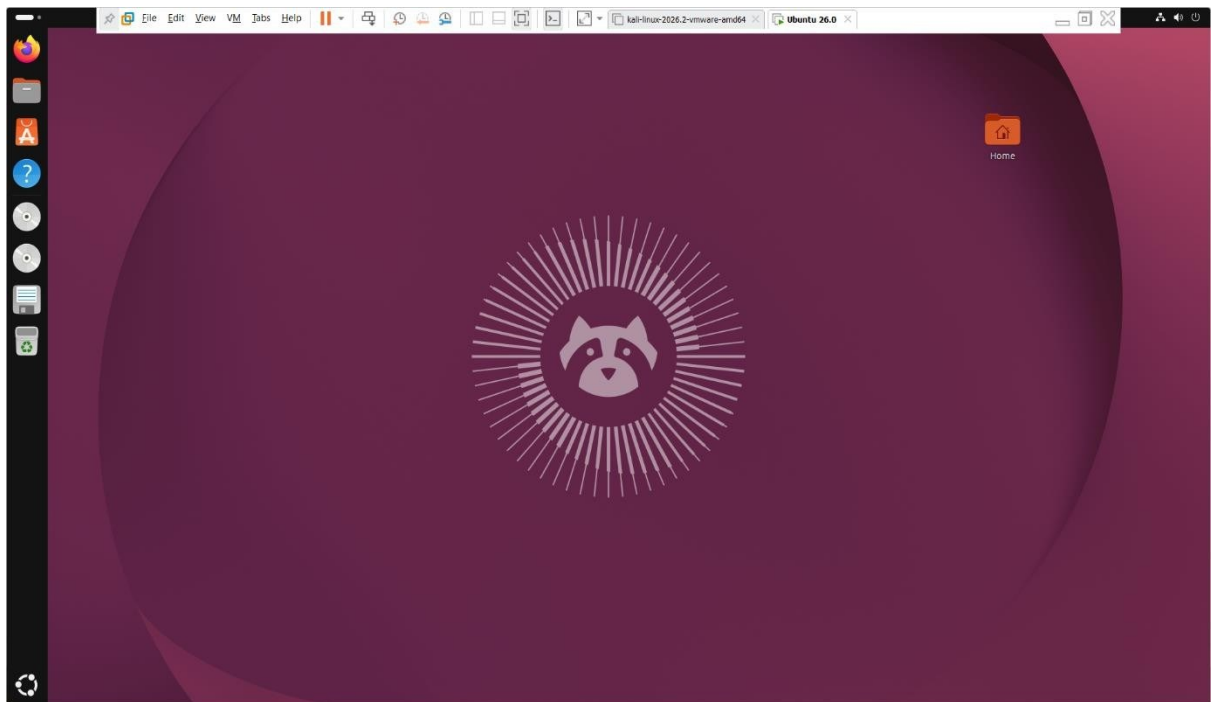
4. Select **Installer disc image file (ISO)**.
5. Browse to the downloaded Ubuntu ISO file.
6. Select Ubuntu as the guest operating system.
7. Enter a suitable virtual machine name, for example:

Ubuntu-VM
8. Select a location where the virtual machine files will be stored.
9. Configure the virtual hard disk.
10. Allocate suitable RAM and processor resources according to the host computer.
11. Select **NAT** as the network connection for a basic setup.
12. Review the configuration.
13. Click **Finish**.

Step 5: Install Ubuntu Inside VMware

1. Power on the Ubuntu virtual machine.
2. The Ubuntu installation screen will appear.
3. Select the required language.
4. Select the keyboard layout.
5. Configure the network connection if required.
6. Select the normal installation option.
7. Select the virtual disk created for the VM.
8. Create the required username and password.
9. Allow Ubuntu to complete the installation.

10. Restart the virtual machine after installation.
11. Log in to the Ubuntu desktop.
12. Verify that Ubuntu is running correctly inside VMware Workstation.



Step 6: Download Kali Linux

1. Open the official Kali Linux download page.
2. Kali provides both **Installer ISO images** and **pre-built VMware virtual machine images**. For this experiment, the pre-built VMware image can be used because it is specifically prepared for VMware.
3. Download the **Kali Linux VMware image**. **Official Kali Linux download:**



Kali Linux Official Downloads

4. The current official Kali download page provides VMware images for x86-64 systems.
5. Save the downloaded .7z file on the host computer.

Step 7: Extract the Kali VMware Image

1. Install/use **7-Zip** or another compatible archive utility.
2. Extract the downloaded Kali VMware archive.
3. After extraction, locate the VMware configuration file with the .vmx extension.
4. Do not delete the other files in the extracted VM folder because the virtual disk and configuration files are required.

Kali's official documentation uses this same process: extract the VMware archive and then open the .vmx file in VMware Workstation.

Step 8: Open Kali Linux in VMware Workstation

1. Open VMware Workstation.
2. Select **Open a Virtual Machine**.
3. Navigate to the extracted Kali Linux VMware folder.
4. Select the .vmx file.
5. Click **Open**.
6. Kali Linux will appear in the VMware Workstation virtual machine list.
7. Review the VM hardware settings if required.
8. Allocate appropriate RAM and processor resources.

9. Configure the network adapter.
10. Select **NAT** for a basic Internet-connected laboratory setup.

Step 9: Start Kali Linux

1. Select the Kali Linux VM in VMware Workstation.
2. Click **Power On**.
3. Wait for Kali Linux to boot.
4. If VMware displays a prompt asking whether the VM was moved or copied, select the appropriate option for your setup.
5. Log in to Kali Linux.
6. Verify that the Kali desktop loads successfully.

For the official pre-built Kali VMware image, Kali currently documents the default login as **kali / kali**. Change the password after first login if required.



Step 10: Configure Ubuntu and Kali Linux

1. Start the Ubuntu virtual machine.
2. Verify that the Ubuntu desktop is working correctly.
3. Start the Kali Linux virtual machine.
4. Verify that the Kali desktop is working correctly.
5. Check the network connection in both operating systems.
6. Adjust RAM, CPU, storage and display settings if necessary.
7. Install or verify the appropriate VMware guest tools where required.
8. Keep the Ubuntu and Kali virtual machines as separate guest environments.

Step 11: Verify Network Connectivity

In Ubuntu, open the Terminal and check the network configuration:

```
ip addr
```

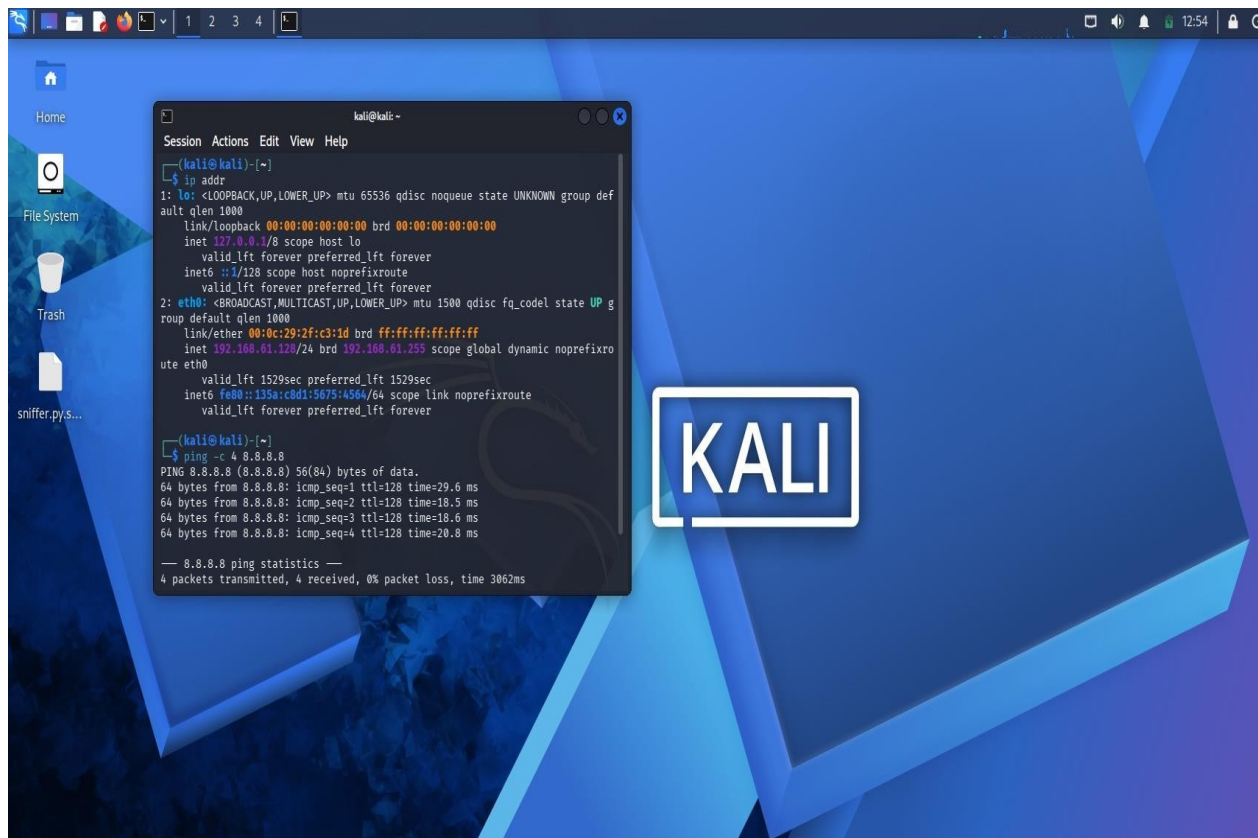
In Kali, open the Terminal and use: `ip`

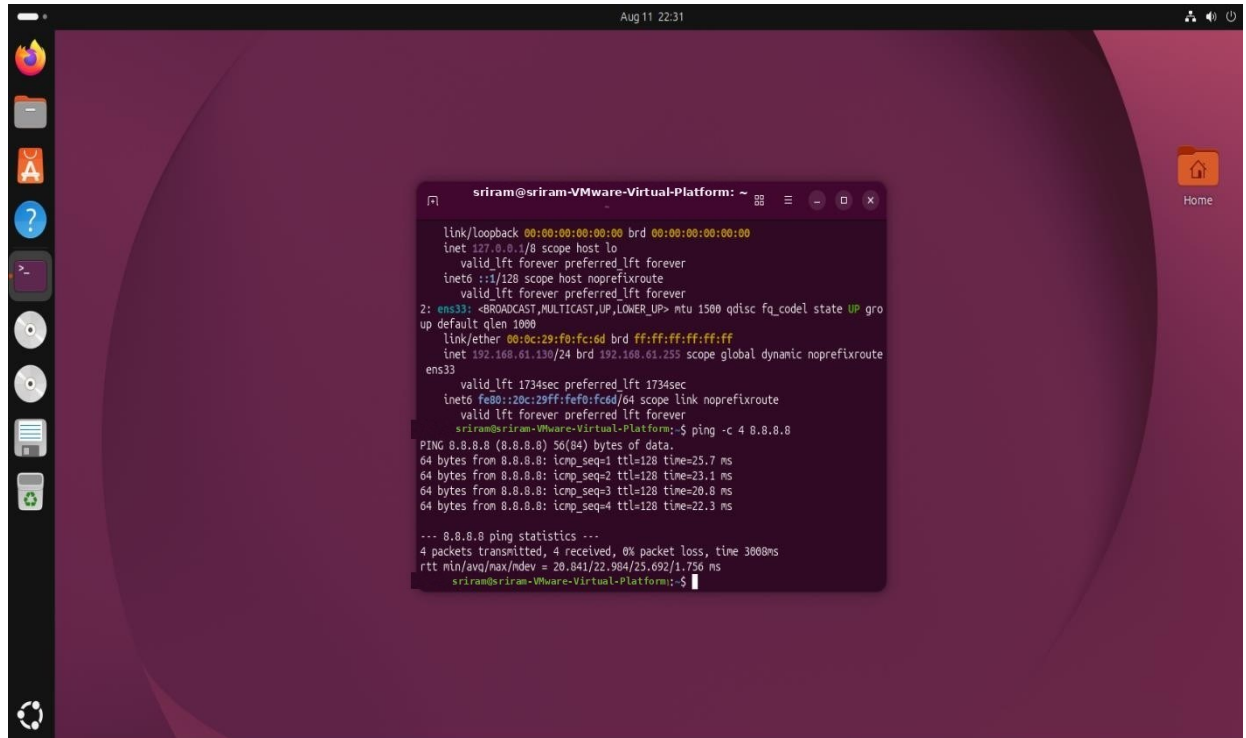
```
addr
```

To test Internet connectivity, use:

```
ping -c 4 8.8.8.8
```

If replies are received, the network connection is working.





```
sriram@sriram-VMware-Virtual-Platform: ~  
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
inet 127.0.0.1/8 scope host lo  
    valid_lft forever preferred_lft forever  
inet6 ::1/128 scope host noprefixroute  
    valid_lft forever preferred_lft forever  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro  
up default qlen 1000  
    link/ether 00:0c:29:f0:fc:6d brd ff:ff:ff:ff:ff:ff  
    inet 192.168.61.130/24 brd 192.168.61.255 scope global dynamic noprefixroute  
    ens33  
        valid_lft 1734sec preferred_lft 1734sec  
    inet6 fe80::20c:29ff:fe0:fc6d/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
sriram@sriram-VMware-Virtual-Platform:~$ ping -c 4 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:  
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=25.7 ms  
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=23.1 ms  
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=20.8 ms  
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=22.3 ms  
--- 8.8.8.8 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3008ms  
rtt min/avg/max/mdev = 20.841/22.984/25.692/1.756 ms  
sriram@sriram-VMware-Virtual-Platform:~$
```

CONCLUSION:

VMware Workstation was successfully installed on the host computer. Ubuntu and Kali Linux were configured as separate virtual machines inside VMware Workstation. Both guest operating systems were successfully started and their basic configuration and network connectivity were verified. Thus, a virtual laboratory environment containing Ubuntu and Kali Linux was established for performing controlled cybersecurity experiments.



Marwadi
University
Marwadi Chandarana Group



FACULTY OF TECHNOLOGY
Department of CE – AI ML DS
01AI0509– Cyber security